Connected Still Life

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Lossfunk

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Abstract

A still life of Conway's Game of Life is a pattern that does not change: every live cell has two or three live neighbours and no dead cell has exactly three. The maximum density of an unconstrained still life approaches $1/2$ asymptotically, and the unconstrained finite-box optimum is known exactly. Requiring the live cells to form a single connected group, under horizontal, vertical, and diagonal adjacency, changes the finite-box optimization problem, because isolated stable pieces can no longer be tiled freely. This problem asks for connected still lifes in an $n \times n$ box with as many live cells as possible, for the box sizes $n = 8, 9, 10, 11, 13, 15, 16, 17, 19, 20, 21, 31, 32$. For small n good solutions are known; for $n \geq 16$ the best-known connected patterns came from ad hoc searches and could be suboptimal. The connected optimum can be close to the unconstrained finite-box optimum for some sizes, so the gap is instance-dependent. Every submission is checked automatically.

Problem

Every cell of the grid is alive (1) or dead (0) and has eight neighbours: the cells directly above, below, left, right, and on the four diagonals. At each time step a live cell survives if it has exactly two or three live neighbours, and a dead cell becomes alive if it has exactly three. A *still life* is a pattern that does not change after a time step, and a *connected* still life is one whose live cells form a single group under the same eight directions. Choose a box size n from the list above and find a connected still life inside an $n \times n$ grid, with everything outside the grid dead, that has as many live cells as possible.

Submission and verification

A submission is a JSON object with the box size `n`, the claimed number of live cells `claimed_cells`, and the $n \times n$ binary `grid`. The verifier checks that

1. the grid is $n \times n$ with entries 0 or 1;
2. every live cell has exactly 2 or 3 live neighbours;
3. no dead cell in the box or in its one-cell exterior ring has exactly 3 live neighbours;
4. all live cells are 8-connected (breadth-first search);
5. the number of live cells equals `claimed_cells`.

Verification takes $O(n^2)$ time. Every valid connected still life for an allowed n is accepted.

Scoring

Submissions are scored by the number of live cells, higher being better, and by the density live cells$/n^2$. Finite optima can exceed 50% because cells near the boundary need fewer neighbours, while the infinite-lattice density is at most $1/2$ (Elkies).

Note added in October 2026

The exact values for all thirteen sizes, and many more, are determined in S. Mahajan, *Extremal Connected Still Lifes in Square Boards*, 2026, https://siddhartha-mahajan.github.io/papers/extremal-connected-still-lifes/.

References

[1] N. D. Elkies, The still-life density problem and its generalizations, *Voronoi's Impact on Modern Science*, Book I, 228–253, 1998; arXiv:math/9905194.

[2] G. Chu and P. J. Stuckey, A complete solution to the maximum density still life problem, *Artificial Intelligence* 184–185 (2012), 1–16.

[3] Connected maximum-density still life, benchmark of the Fourth Answer Set Programming Competition, 2013, https://www.mat.unical.it/aspcomp2013/ConnectedMaximumDensityStillLife.